Remote sensing as a novel approach for addressing micro plastic detection in the environment

Despite all the environmental discussions on the dangers and hazards of plastic materials, plastic production worldwide is growing at a rate of about 5% per year. Plastic waste is recognized as one of the main threats for the environment as the presence of plastic litters affects the whole environment, from the atmosphere to the marine domain (Simoneit et al., 2005). Plastic waste is easily transported by wind and water due to its low density, complicating plastic waste management and plastic pollution is considered to a major threat to our planet (Boodhoo *et al.,* 2023). Mismanagement of plastics disposal causes their spread in the environments where they are brittle and fragmented by the effect of heat, UV rays, winds, waves, forming microplastics (MPs), defined as plastic particles less than 5 mm in diameter.

According to Rednikin *et al (*2024), and Longyi *et al* (2022), depending on their origin, MPs can be classified into two categories namely primary and secondary MPs. Primary MPs are intentionally produced as micro-sized particles for specific purposes such as personal care products and they can enter the environment through spills or any other incidental release during production or transportation of the material. Secondary MPs originate from degradation, aging, and fragmentation of large particles, typically caused by physical abrasion and/or exposure to ultraviolet radiation and temperature variation and they make up most MPs circulating in the environment. He particularly emphasizes about airborne microplastics, expressing that they are abundant in both urban and natural environments, including water bodies and glaciers, as particles can travel long distances.

Plastic particles, especially those < 10 μm, entering the human body through inhalation or ingestion have been shown to cause serious adverse health effects. Therefore, environmental pollution from MPs in the air is a matter of growing concern. However, Rednikin et al (2024) further points out that the airborne microplastics (AMPs) have not been sufficiently explored yet, since the current studies mainly focus on the analysis of distribution and behavior of MPs in water bodies, and hence the approaches to improve sampling techniques and develop standardized methods to assess airborne and terrestrial microplastics are still far from being perfect. Longyi *et al (*2022) also highlights the lack of standard methods for the identification and quantification of atmospheric MPs and urgent need for integration and critical analysis of various research units for new research directions. Singh (2024) also stresses the need of standardized methods in sampling and measuring MPs in the environment ensuring the reproducibility and comparability of results.

Remote sensing technology, on the other hand, is currently being used in various environmental monitoring applications. In recent years, remote sensing has shown great potential for monitoring marine plastic litter using surveillance cameras, drones, and satellite imagery. Furthermore, high resolution satellites with multispectral bands and specific data models also have shown promise in detecting suspected plastic accumulation zones even though standardized monitoring methods for plastic waste detection has not yet been developed (Blume, et al, 2023). Hence, it would be reasonable to evaluate the applicability of remote sensing technology in detecting the type, quantity, and the behavior of MPs in the environment.

Bentley, J. (2019) explains that plastics are hydrocarbon composites materials that share common absorption wavelength at 1730 and 2310 nanometers independently of the plastic type and hence it is possible to use hydrocarbon detection indices to detect plastic components. (Hydrocarbon Detection Indices are metrics used to assess the presence and concentration of hydrocarbons in a specific area or environment (OpenAI, personal communication, February 5, 2025). According to Wienke, D., 1995, plastics have unique inherent optical characteristics in the near infrared (NIR) to shortwave infrared (SWIR) spectrum that have been used in automated optical sorting of waste in the recycling industry in which reflectance in the visible region to NIR region (400 to 900 nm) has been measured on selected pieces of macro plastic garbage. However, accordingly there is a rising interest in the potential applications of remote sensing in detecting hydrocarbons such as oil, plastics and methane utilizing the absorption features in the ultra-violet to longwave infrared spectrum.

Mukonza, (2022) highlights that there is increasing availability of free big geospatial data from satellite sensors for potentially monitoring plastics as an alternative to high-cost sampling and monitoring systems currently being used for airborne microplastic research in the laboratory. He proposes that this could be a possible solution to these challenges by minimizing the fieldwork, costs and sampling time. Bentley, J. (2019) also suggests a similar proposal that emerging technologies such as remote sensing as a cost-effective and capable approach of monitoring micro plastics in the environment. Waqas, et al (2023) supports the fact that remote sensing data has proved its potential and usability in monitoring environmental issues, especially in identifying, quantifying, and detecting plastic litter in the terrestrial and marine environment.

Bentley, J. (2019) further reveals about a few attempts made in using hyperspectral imagery for detecting varieties of polymers. Discutido et al (2024) argues that spectral mixture modeling is useful in estimating pixel abundance of unique optically active components. Bentley, J. (2019) further emphasizes that spectral unmixing has proved to be a powerful method for plastic detection. (Spectral unmixing is a technique used in remote sensing and imaging spectroscopy to separate the contributions of different materials or substances within a single pixel of an image. Since each pixel in a hyperspectral or multispectral image can contain a mixture of different materials, spectral unmixing helps to estimate the abundance of each material present (OpenAI, personal communication, February 7, 2025). Plastic debris, whether they are floating or on the ground, are usually smaller than the pixel size and are therefore not large enough to form pure pixels. However, the interpolation can be used to estimate a pure plastic pixel even though it induces uncertainties regarding plastic presence at subpixel scale.

Discutido *et al,* (2024) addresses the escalating environmental concern of airborne microplastics by developing a low-cost detection machine employing infrared sensors and this research has advanced the capacity to detect airborne microplastics. Their results suggest that sensors with the appropriate spectral and spatial resolution could potentially provide a resource to assess the location and extent of aggregations of hydrocarbons like plastics in the natural environment. Additionally, they suggest including remote sensing tools as part of integrated multi-disciplinary strategy to quantify plastics in the environment. According to Discutido *et al,* (2024) it is appropriate to evaluate the potential in the mapping of plastics, even though remote sensing of plastic debris from a distance is still in the early phases.

Incidentally, a team of researchers at the University of Michigan has developed a new method to map the concentration of ocean microplastics over the globe using data from eight microsatellites that are part of NASA's Cyclone Global Navigation Satellite System. This research is the first to map ocean microplastics over a large area of the ocean and is the first to map concentrations at a high temporal resolution, revealing seasonal variations in microplastic concentrations (https://www.earthdata.nasa.gov). And, researchers from the Remote Sensing Laboratory of the National Technical University of Athens (NTUA) and the National and Kapodistrian University of Athens (NKUA) have recently developed a marine debris data set based on the multispectral data from the Copernicus program's Sentinel-2 mission (Blume, et al, 2023).

Microplastics are minute particles that are not visible to human eye. The resolutions of the most currently available satellite sensor types are not adequate enough for capturing those particles from distance. And also, Discutido *et al,* (2024) highlights that the surface texture and characteristics of MPs are rapidly changing due to different factors such as weathering processes. Furthermore, absence of developed models to detect and quantify the airborne micro plastics is still a challenge.

Remote sensing has already been applied in some preliminary studies in plastic litter detection such as detection of marine micro plastics (Blume et al, 2023). Amongst the above possibilities of applications of RS in the field of plastic litter detection, there is a need for critically analysing these feasibilities in microplastics detection and assessment in the air, water and land individually. The unique absorption values and reflectance values of plastics and the possibility to use spectral unmixing algorithms in GIS technology are positive gears towards the applicability of remote sensing technologies in micro plastic detection in the environment. Furthermore, Chabot & Marteinson (2024) suggest that experimenting with image acquisition at much lower altitudes and finer spatial and spectral resolutions, which can be conveniently achieved using drones equipped with high-precision hyperspectral sensors would be much beneficial in this ground. Hence, despite the other limitations, the potential of applying remote sensing technology in microplastic detection in the environment, integrating machine learning algorithms together is stressed to be harnessed through this review.

Disclaimer: Some portions of this article were generated with the assistance of ChatGPT, an AI language model. The use of AI tools has been carefully managed, and all content has been thoroughly reviewed by the author for accuracy and appropriateness.

References

1. Bentley, J. (2019). Detecting Ocean Microplastics with Remote Sensing in the Near-Infrared: A Feasibility Study. In BSU Honors Program Theses and Projects. Bridgewater State University, Item 309. https://vc.bridgew.edu/honors_proj/309

2. Blume, S., Franke, J., Garaba, S., Giang, P., Mathis, J., Ortwig, N., & Ziegler, S. (2023). *Advances in remote sensing of plastic waste*. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH. https://www.giz.de/de/downloads/giz-2023-en-advances-in-remote-sensing-of-plastic-waste.pdf

3. Boodhoo, K. (2023). Holistic approach to tackle (Micro) plastic pollution: The case of Mauritius. Global Journal of Science Frontier Research: H Environment and Earth Science, 23(1), 33–45. https://globaljournals.org/GJSFR_Volume23/5-Holistic-Approach-to-Tackle-Micro-Plastic-Pollution.pdf

4. Chabot, D., & Marteinson, S. C. (2024). Aerial Remote Sensing of Aquatic Microplastic Pollution: The State of the Science and How to Move It Forward. *Microplastics*, *3*(4), 685-695. https://doi.org/10.3390/microplastics3040042

5. Discutido, C. D. J., Sia, A. A. L. R., and Dela Cruz, C. M. (2024). Near-Infrared Light and OpenCV as Components for Low-Cost Airborne Microplastic Detection Machine, ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., X-5-2024, 57–62, https://doi.org/10.5194/isprs-annals-X-5-2024-57-2024

6. Dubucq D., Jatiault R., Boymond, P, Rusquet.A, (2021). Remote sensing detection of plastic waste: recent improvements and remaining challenges. Proceedings of SPIE, the International Society for Optical Engineering, Earth Resources and Environmental Remote Applications Sensing/GIS XI, 11534, pp.2574026. 10.1117/12.2574026hal-04498670, https://hal.science/hal-04498670v1

7. Mukonza, S. S., Chiang, J. L. (2022). Satellite sensors as an emerging technique for monitoring macro- and microplastics in aquatic ecosystems. *Water Emerg. Contam. Nanoplastics* , *1*, 17. http://dx.doi.org/10.20517/wecn.2022.12

8. Rednikin, A. R., Frank, Y. A., Rozhin, A. O., Vorobiev, D. S., & Fakhrullin, R. F. (2024). Airborne Microplastics: Challenges, Prospects, and Experimental Approaches. *Atmosphere, 15*(11), 1380. https://doi.org/10.3390/atmos15111380

9. Simoneit, B. R., Medeiros, P. M., & Didyk, B. M. (2005). Combustion products of plastics as indicators forrefuse burning in the atmosphere. Environmental science & technology, 39(18), 6961-6970, https://www.researchgate.net/publication/7562860_Combustion_Products_of_Plastics_as_Indicators_for_Refuse_Burning_in_the_Atmosphere, DOI: 10.1021/es050767x

10. Singh, B.P. (2024). Understanding the dynamics and implications of airborne microplastics in atmosphere, Air Quality, Atmosphere & Health. https://doi.org/10.1007/s11869-024-01593-4

11. Shao, L., Li, Y., Jones, T., Santosh,M., Liu, P., Zhang,M., Xu,L., Li,W., Lu,J., Yang, C.X., Zhang,D., Feng,X., BéruBé, K. (2022). Airborne microplastics: A review of current perspectives and environmental implications, Journal of Cleaner Production, Volume 347,2022,131048,ISSN 0959-6526, https://doi.org/10.1016/j.jclepro.2022.131048.

12. Waqas, M., Wong, M. S., Stocchino, A., Abbas, S., Hafeez, S., & Zhu, R. (2023). Marine plastic pollution detection and identification by using remote sensing-meta analysis. Marine Pollution Bulletin, 197, 115746. https://doi.org/10.1016/j.marpolbul.2023.115746

13. Wienke, D., Van den Broek, W. H. A. M., & Buydens, L. (1995). Identification of plastics among non-plastics in mixed waste by remote sensing near-infrared imaging spectroscopy. 2. Multivariate image rank analysis for rapid classification. *Analytical Chemistry*, *67*(20), 3760-3766. 10.1021/ac00116a023.

14. NASA, Tracking ocean plastic from space, https://www.earthdata.nasa.gov/news/feature-articles/tracking-ocean-plastic-from-space, accessed on 02nd February, 2025

15. Open AI. (2025). ChatGPT (February 5, 7) [Spectral unmixing, Hydrocarbon detection indices]. https://chat.openai.com/chat

Justification

Initially in the assignment 1, I started the paper with several aspects on the subject area in which I was curious in by referring several research questions. After the feedback received for assignment 1, I narrowed down the topic into one specific area, ie. assessing the potential of remote sensing technology in detecting atmospheric micro plastics. Moreover, I continued writing with the suggested paragraph as a basis for the second assignment. There, I specifically addressed only one aspect as the thesis statement.

After the feedback received on the second assignment, I broadened the scope of the review to address micro plastics in the environment instead of narrowing to air borne microplastics. The reason was the limited studies available on the micro plastics in the atmosphere. And also, I tried to rephrase the sentences to my words as possible to emphasize a clearer idea on the topics while making several changes on the sentences and replacing them with more appropriate ones based on several new literature.

In the introductory section, I added a few paragraphs to highlight the characteristics of micro plastics specifically to highlight the importance of addressing the scenario. In the body paragraphs I addressed some specific points that I could find on the literature such as the spectral behavior of plastic materials, application instances of RS in plastic litter detection in different environments (water, land and air) and some highlights on spectral mixing and unmixing algorithms related to this area as suggested in the feedback. However, due to space limitations I could not address these areas in detail.

Mor importantly, I changed the title of my literature review as “Remote sensing as a novel approach for addressing micro plastic detection in the environment” by broadening the scope of the research.

Moreover, I took some inputs from the comments suggested by the reviewer of my second paper. Accordingly, I addressed the limitations of the RS technology in this area to make it a two-sided approach.

Personal reflection

The entire course on scientific writing is significantly important to me as a prospective researcher in the future. This exercise was a turning point of my scientific writing career as it guided me towards a more specific path of communicating research findings. In overall I could say that I was able to grab the proper way of scientific communication through this course.

In the first writing, it was rather challenging for me to finding and writing a thesis statement as there was no proper guidance given on that. However, after having the personal feedback on the first writing I got it resolved. I would like to emphasize the significance of giving personal feedback for each and ever person as if directly affects the way of their writing. For me, all the feedback provided personally were invaluable. I reflected on each and every opinion pointed out on my writings. I have no words to thank the instructors for their individual concern for us.

If I highlight some notable points, I grasped from this exercise, how to narrow down into a particular scope in a given subject area, how to streamline research questions, and how to focus on the scope of the research are of great concern. Additionally, I could learn to accurately cite a particular article within a text as well as in the references.

Even though I have little intension on applying this particular review for my master thesis due to some other interests, this literature review made me awakened on a specific topic to reconsider in my future studies for further research.